

Overall verdict: the roadmap is solid and data-grounded. But it has 4 specific gaps that your EDA directly exposes.

Module 1 — Spatial hotspots: approved, with one fix

The DBSCAN plan is right. But your EDA shows the K-distance plot is already done using Euclidean distance on raw lat/lon degrees. This is technically wrong — a degree of longitude in Bangalore is about 95km, but a degree of latitude is ~111km, so the coordinates are not isotropic. You should convert to radians and use `metric='haversine'` in DBSCAN. With 298,450 points this matters for epsilon selection. Set `algorithm='ball_tree'` alongside it. The K-distance plot needs to be rerun after this correction to pick the right epsilon.

Module 2 — Temporal patterns: keep, with a critical reframe

The 5 AM peak is the most important finding in your entire EDA and the roadmap handles it cautiously but correctly by calling them "detection hours." Your EDA confirms the pattern clearly: hours 2–6 AM dominate, with a secondary evening cluster from 19–23. This almost certainly reflects officer patrol shifts, not actual violation peaks. The reframe to tell judges: *"violation detection time ≠ violation occurrence time — our temporal analysis reveals patrol shift patterns, which is itself operationally useful for scheduling."* That reframe turns a data quality limitation into an insight. Don't hide it.

One thing missing from Module 2: your EDA shows Sunday and Saturday are the top 2 days. But you also have only 5 months of data (Nov–Apr, with April being partial). You should flag to judges that seasonal patterns are incomplete — you're seeing a winter–spring window only, not a full annual cycle.

Module 3 — Congestion risk engine: one important addition

The severity weights are good. The junction multiplier is good. The multi-violation bonus is good. But your EDA revealed something the roadmap didn't fully exploit: the actual named junctions. `BTP051 - Safina Plaza` has 15,449 violations. `BTP082 - KR Market` has 11,538. These are not a binary junction flag — they're graded by volume. The roadmap says `junction_flag = 1 if junction != "No Junction"` but you could do better: use `junction_violation_count` (normalized) as the junction weight instead of a flat binary. Zones near high-volume junctions score higher than zones near low-volume ones. This is entirely data-supported and more defensible than a flat flag.

Module 4 — XGBoost classifier: one gap

The classification framing (Low/Medium/High/Critical) is the right call over regression. But the roadmap's feature list doesn't include `vehicle_type`, which your EDA shows is highly informative — SCOOTER (94,856) and CAR (88,870) dominate, but TANKER, HGV, and LORRY parked illegally have a fundamentally higher congestion impact than a scooter. Vehicle type should be in the severity scoring (Module 3) AND the XGBoost feature table. A TANKER violating in a main road is categorically different from a scooter with wrong parking.

The roadmap's `severity_weight` table only accounts for violation type, not vehicle type. Add a `vehicle_size_weight` multiplier: two-wheelers = 1.0, cars = 1.5, LGV/tempo = 2.5, HGV/lorry/tanker = 4.0. This is defensible from first principles (road space consumed) and adds a second data-native dimension to your risk score.

Module 5 — Patrol allocation: approved as-is

No issues. The output format (zone, predicted risk, suggested window) is exactly right for judges.

Module 6 — Impact simulator: keep, but scope it tightly

This is the right differentiator. The only risk is scope creep. Keep it to one slider: "% reduction in violations at Zone X" → show the change in composite priority score. Don't try to model traffic flow — you've already pre-empted that with your surrogate model disclaimer. The simulator should be a heuristic calculator, not a simulation, and you should say so explicitly.

One gap the roadmap missed entirely: the `validation_status` issue

Your EDA shows 125,254 records (42%) have `validation_status = NaN`, and 49,754 are `rejected`. The roadmap says to use all data but doesn't address this. You have three options and need to pick one consciously: use only `approved` (115,400 records, cleanest but smallest), use `approved + NaN` (most records, reasonable since NaN likely means not-yet-reviewed rather than invalid), or use everything. I'd recommend `approved + NaN` and exclude `rejected` and `duplicate`. Document this choice explicitly — judges will ask.

One gap regarding the DBSCAN notebook ending

Your EDA ends with the K-distance plot setup but no DBSCAN execution output. That means cluster IDs don't exist yet. Everything downstream — persistence scoring, XGBoost features, patrol zones — depends on this running first. This is your immediate next step before anything else.

In summary: the roadmap is well-grounded in your EDA findings. The four things to act on are the haversine DBSCAN fix, the graded junction weight (not binary), adding `vehicle_type` to the severity multiplier, and deciding your `validation_status` filter before Module 1 runs. Everything else in the roadmap stands.